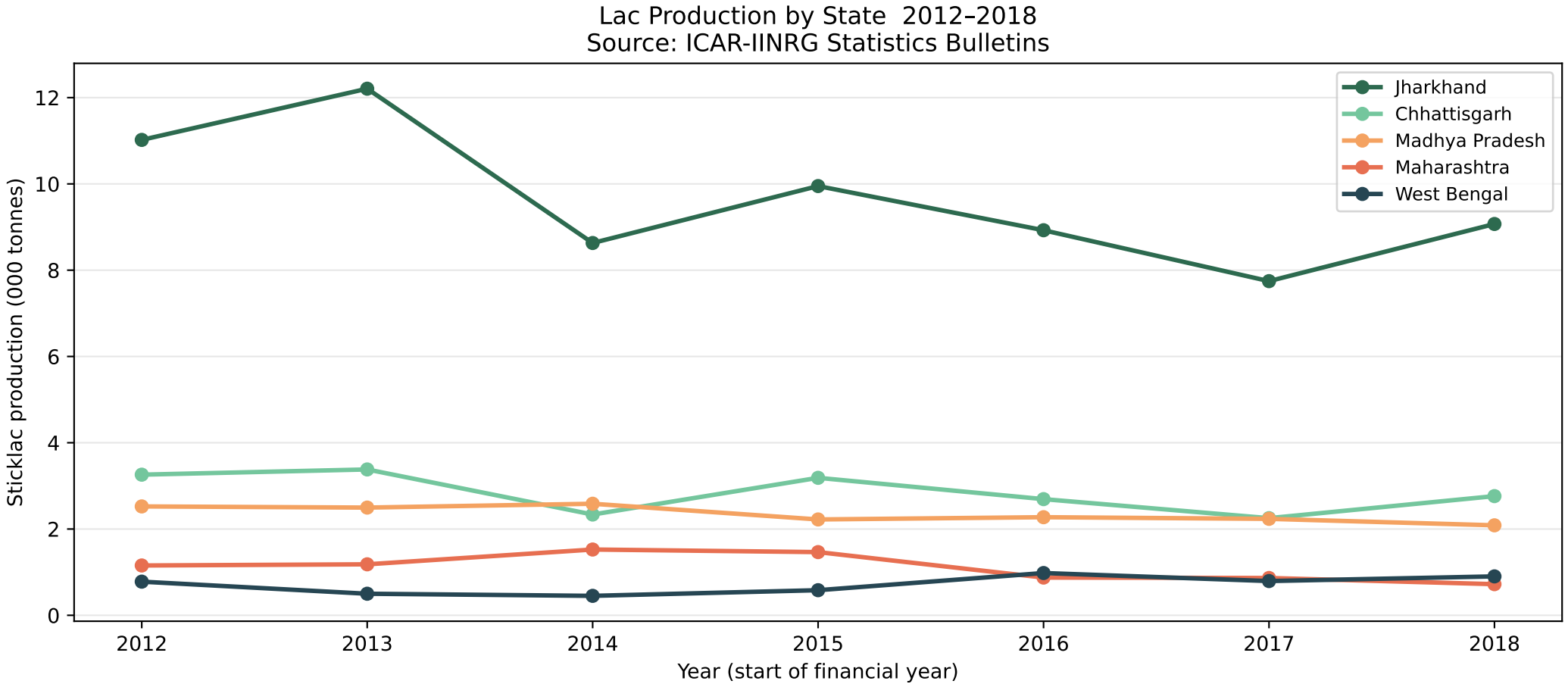
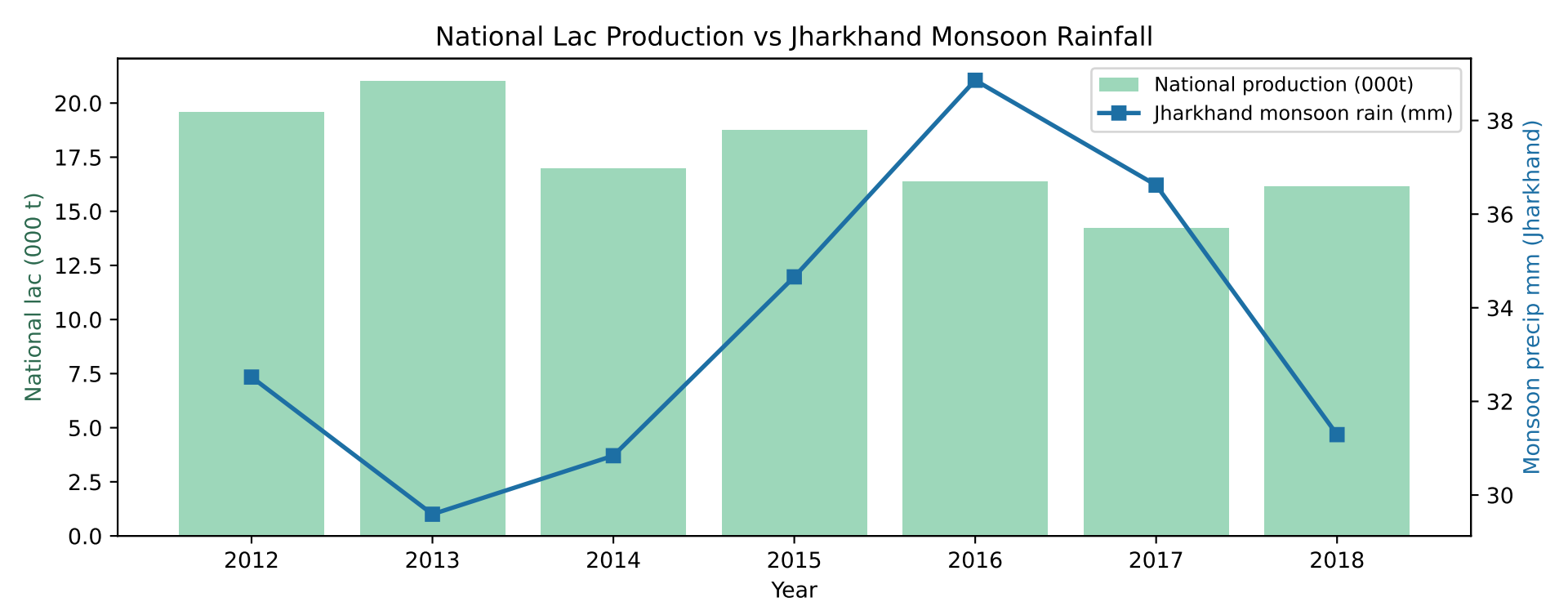
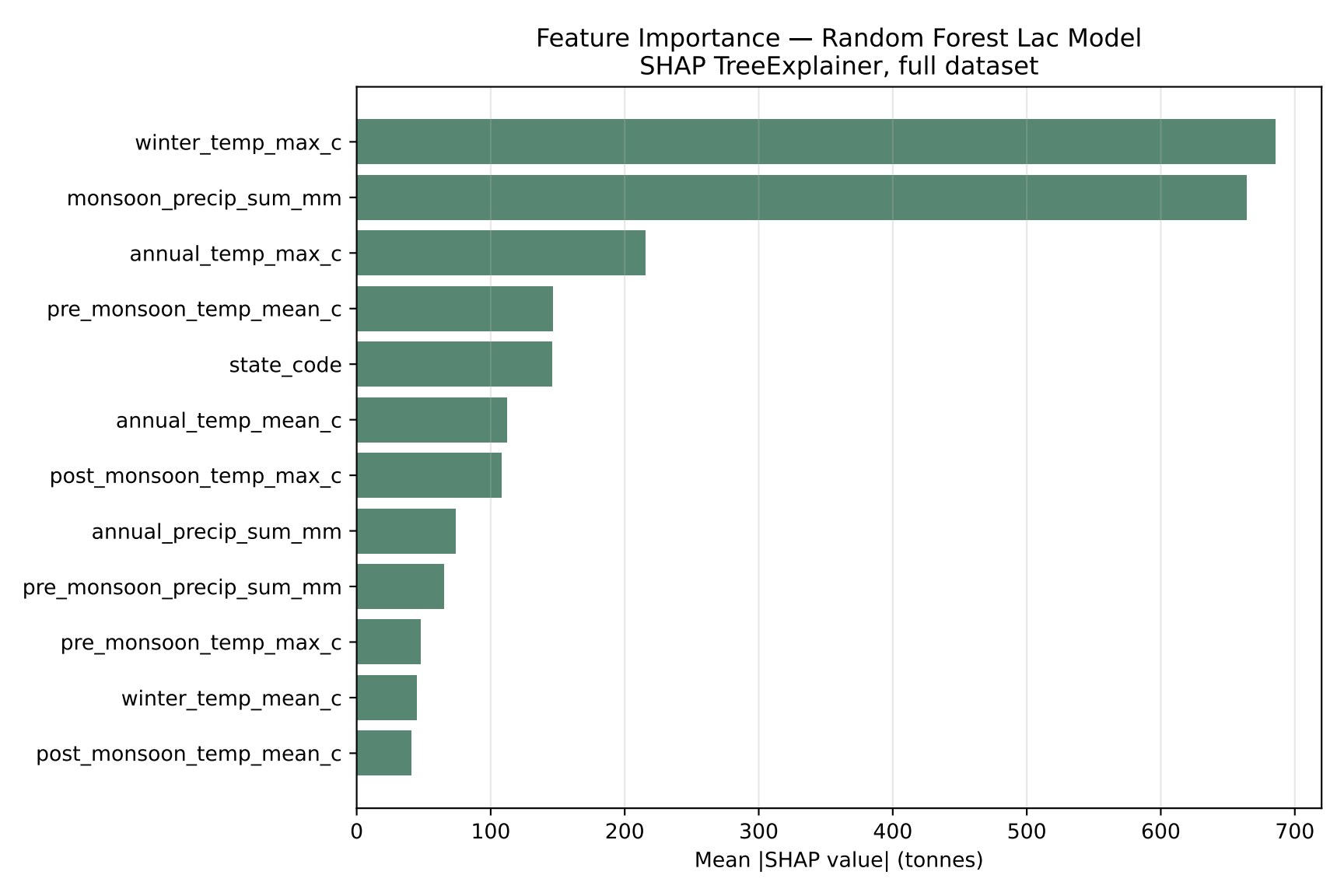
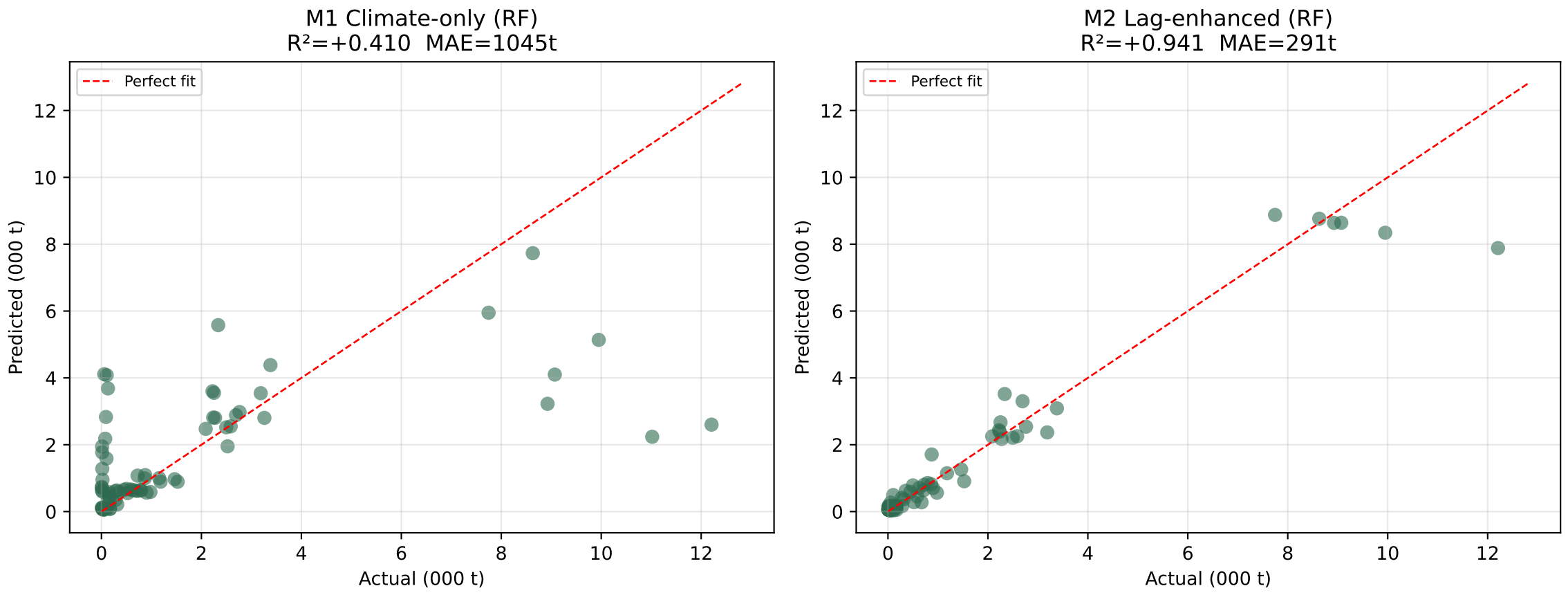
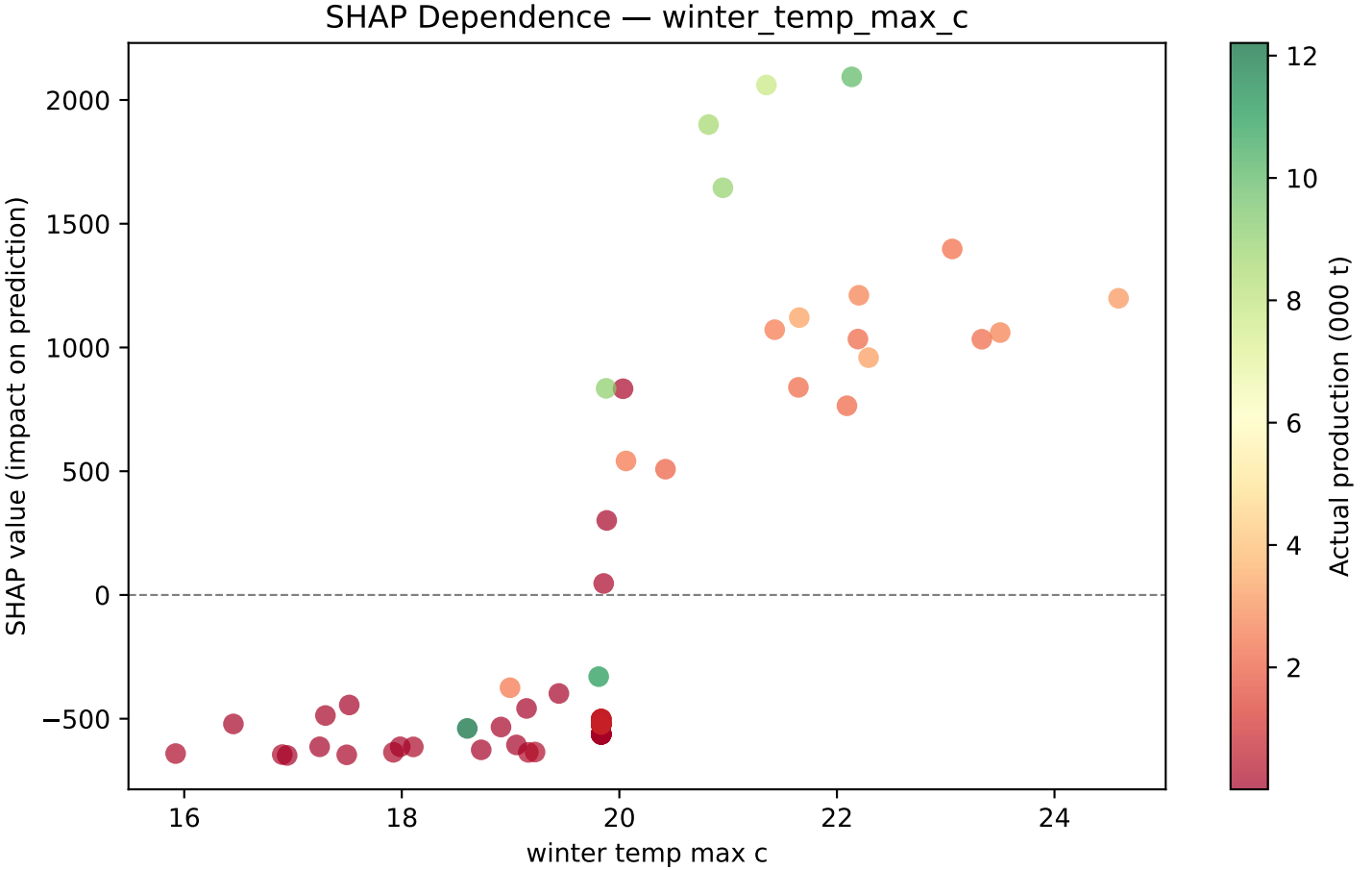

Lac Production by State 2012–2018
Source: ICAR-IINRG Statistics Bulletins
Sticklac production (000 tonnes)
0
2
4
6
8
10
12
2012
2013
2014
2015
2016
2017
2018
Year (start of financial year)
Jharkhand
Chhattisgarh
Madhya Pradesh
Maharashtra
West Bengal


National Lac Production vs Jharkhand Monsoon Rainfall
National production (000t)
Jharkhand monsoon rain (mm)
National lac (000 t)
Monsoon precip mm (Jharkhand)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
30
32
34
36
38
2012
2013
2014
2015
2016
2017
2018
Year


Feature Importance — Random Forest Lac Model
SHAP TreeExplainer, full dataset
winter_temp_max_c
monsoon_precip_sum_mm
annual_temp_max_c
pre_monsoon_temp_mean_c
state_code
annual_temp_mean_c
post_monsoon_temp_max_c
annual_precip_sum_mm
pre_monsoon_precip_sum_mm
pre_monsoon_temp_max_c
winter_temp_mean_c
post_monsoon_temp_mean_c
0
100
200
300
400
500
600
700
Mean |SHAP value| (tonnes)


LOYO Cross-Validation: Actual vs Predicted Lac Production
M1 Climate-only (RF)
R²=+0.410 MAE=1045t
M2 Lag-enhanced (RF)
R²=+0.941 MAE=291t
Perfect fit
Predicted (000 t)
Actual (000 t)
0
2
4
6
8
10
12


SHAP Dependence — winter_temp_max_c
SHAP value (impact on prediction)
2000
1500
1000
500
0
−500
16
18
20
22
24
winter temp max c
Actual production (000 t)
12
10
8
6
4
2